

Building Consensus in the Circular Economy: A Transdisciplinary Framework for Developing Collaborative Decision-Making Tools

Mohamed ELNOURANI¹ and Anna Öhrwall RÖNNBÄCK

Department of Social Sciences, Technology and Arts Luleå University of Technology, Sweden

Abstract. The transition towards a circular economy requires an overarching perspective that recognises the dynamic and interdisciplinary nature of our current economic growth landscape. Circular economy inherently involves numerous stakeholders across the product life cycle. To establish efficient circular economy practices among these actors, this study suggests a novel framework for developing collaborative and interdisciplinary decision-making tools. By looking into relevant literature, organising a workshop, and analysing standard tools used in a circular economy like KPIs, risk analysis, Cost-Benefit Analysis, etc., we managed to capture the multidisciplinary challenges and dynamics faced by stakeholders of the circular business model. The contribution of this paper is the development of a framework that bridges requirement management techniques from product development practices with circular economy principles to facilitate effective decision-making processes. The framework effectively balances diverse stakeholder requirements, addressing uncertainties and multi-ownership challenges through product life cycles. This framework may be used to validate existing tools used by businesses and systematically develop new ones when needed. By facilitating collaboration around the circular economy, this framework not only reduces the environmental impact of economic growth but also encourages society to move towards more collective efforts to achieve sustainability. Finally, this article highlights the importance of a transdisciplinary approach in a systematic and effective transition to a circular economy.

Keywords. Circular economy, Decision-Making tools, Collaboration, Requirement management, Transdisciplinary engineering

Introduction

Addressing global resource scarcity demands a shift towards a circular economy, moving away from the linear "make-use-dispose" model associated with environmental degradation and resource overuse. The linear economic model has caused significant harm to the environment, depleted resources, and increased greenhouse gas emissions. The circular economy, a response to these challenges, aims to break the link between economic growth and resource consumption by emphasising recycling, reusing, and

¹ Corresponding Author, Email: mohamed.elnourani@ltu.se.

extending the lifespan of products [1]. Although we now have the technology to recycle, reuse, and extend the life of products, which shows promise for greatly reducing waste and resource use, there is a big missing piece when it comes to the non-technical aspects needed to make this potential a reality. Studies show that the main product's environmental impact is decided during the design phase, but the adoption of a circular economy is often held back by factors other than just technology [2]. These nontechnical barriers, significant and urgent, involve a wide range of challenges in business models, supply chain, and regulations, information sharing and shareholder collaboration, forming significant encounters to the implementation of circular practices [3]. Moreover, Conflicts of interest arise when the goals or benefits sought by one party are at odds with those of another, potentially leading to competition rather than cooperation. A singular viewpoint, where stakeholders focus solely on their segment of the product lifecycle without considering broader implications, can limit the effectiveness of circular strategies. Divergent considerations and different conceptions of value, where what is valuable to one stakeholder might be considered waste or liability by another, complicate decision-making processes and collaborative efforts [2]. The circular economy's collaborative nature requires that decision-support tools not only manage data and information flow but also address the nontechnological barriers to collaboration. These barriers include aligning disparate business models, navigating regulatory landscapes, and fostering mutual trust and understanding among participants [4].

Recognising these needs, this paper's framework seeks to guide organisations in creating or refining decision-making tools, laying the groundwork for enhanced collaboration and a successful circular transition. This framework synthesises insights from diverse scientific fields, such as business, organisational behaviour, data management, and design, showcasing a transdisciplinary approach that integrates knowledge across disciplines. It addresses the multifaceted challenges of circular economy collaboration by providing a structured methodology for tool development that considers the entire product lifecycle, the various stakeholders involved, and the intricate web of interactions that define the circular economy. The study's methodology integrated theoretical analysis and empirical data to develop a decision-making tool framework for the circular economy. It began with a literature review to understand transdisciplinary approaches and decision-making in this context, aiming to categorise crucial tools and demonstrate the integration of diverse perspectives. Insights from this review informed a survey identifying prevalent tool categories, complemented by a workshop with tool manufacturers, SME technology users, and stakeholders in recycling programs. This workshop identified key challenges and opportunities in circular economy practices, with feedback used to map out actor roles and relationships.

1. Theoretical framework

1.1 Decision-making tools in the context of circular economy

Decision-making tools are crucial for navigating the circular economy, enabling sustainable production and consumption. These include Assessment and Analytical Tools (Life Cycle Assessment, Material Flow Analysis, Social Life Cycle Assessment), Strategic and Planning Tools (Circular Business Model Canvas, Value Hill model), and others designed to mitigate information asymmetry and foster collaboration across organisations [4]. By evaluating environmental, economic, and social impacts, they

guide strategic alignment and simplify management of circular initiatives [5]. Moreover, Operational tools streamline processes and enhance cross-organizational compatibility, while Compliance and Reporting Tools ensure adherence to sustainability standards [6]. Innovation and Design Tools promote sustainable design, and Stakeholder Engagement Tools facilitate dialogue and knowledge sharing [7]. Technology and Digitalization Tools (Blockchain, IoT, AI) and Financial Tools (Cost-Benefit Analysis, Return on Investment analysis) overcome information asymmetry and support economic alignment in circular economy projects [8]. However, many tools initially developed for linear models face limitations in the circular context, including single organization focus and data gaps. Challenges like data availability and system integration complexity necessitate tool adaptation for broader lifecycle perspectives and collaborative data sharing to enhance their effectiveness in the circular economy [2].

1.2 Decision-making and collaboration

The growing trends of collaborative decision-making among organisations need to integrate social and technical aspects to support diverse perspectives and objectives [9]. This transition necessitates tools that enhance collaboration, aligning with the paradigm shift towards accommodating diverse perspectives and knowledge bases in decision-making processes highlighted by Zaraté [10]. The development of these tools requires an understanding of the cooperative decision-making model, which considers objectives, processes, and environmental factors, offering a framework for creating tools that support distributed cognitive processes in collaboration [11]. Furthermore, adaptability is crucial in decision-making tools to support various cooperation forms, including complementary, interdependent, and negotiated cooperation, facilitating seamless communication and information sharing [12]. By integrating technological and non-technical aspects of decision-making, these tools can bridge the gap in modern organisational decision-making, addressing the complex nature of collaborative processes. This emphasises the need for tools that not only facilitate but also optimise collaborative decision-making, incorporating meta-planning, negotiation, and evaluation within a cooperative context [13].

1.3 Development frameworks for decision-making tools

The development of decision-making tools is guided by various frameworks and methodologies to ensure they are user-centred, efficient, and responsive to users' needs. For instance, Agile development, highlighted for its iterative progress and adaptability, suits decision-making tools by allowing for flexibility and ongoing collaboration [14]. Design Thinking, which proceeds through phases of empathy, definition, ideation, prototyping, and testing, is tailored for creating solutions that resonate with user needs, making it ideal for decision-making tool development [15]. User-Centered Design places users at the core of the development process, ensuring tools meet user preferences and requirements. Systems Thinking, beneficial for tools addressing complex interconnected systems issues, emphasises understanding systemic complexities and interrelationships [15]. Moreover, Data-Driven Design leverages analytics and user feedback to align tools with actual user behaviours and preferences. The tool development process encompasses needs assessment, requirement gathering, conceptualisation, prototyping, user testing, and continuous improvement, ensuring tools are not only effectively designed but also remain relevant over time by adapting to changing needs and technological advance [16].

The transition towards a circular economy is fraught with complexities that necessitate a coordinated effort across various organisational and stakeholder dimensions. Challenges in harmonising disparate business models, overcoming information asymmetry, and achieving technological interoperability underscore the difficulty of aligning diverse value propositions and operational cultures towards shared circular transition [6, 17]. The situation is further complicated by regulatory discrepancies and the need for economic and financial alignment, emphasising the importance of shared investment models and cultural convergence to facilitate collaborative dynamics and process synchronisation. Additionally, the challenge of scaling circular economy initiatives from concept to widespread application highlights the need for a nuanced approach that can accommodate variations in scale, scope, and resources [2]. At the core of addressing these hurdles is effective stakeholder management, which requires balancing strategic, ethical, and practical considerations. Guided by principles of stakeholder saliency, the imperative of engaging with stakeholders through transparent communication and ethical consideration becomes evident. This engagement ensures the alignment of stakeholder expectations with organisational objectives, fostering an environment where instrumental and normative considerations are balanced, thereby maintaining long-term legitimacy and support [18].

1.4 Interdisciplinary integration to support collaboration

Interdisciplinary collaboration enhances decision-making in sustainability, integrating diverse disciplines for a comprehensive understanding of complex systems. This approach, emphasising systems-informed perspectives and adaptive management, facilitates collaboration and sustainable decision-making, supported by literature such as Pricope et al. [19] and Daniel, et al. [20]. The challenges of interdisciplinary research, including coordination and integration across scales, necessitate collaborative platforms and tools but also offer opportunities for innovative sustainability practices within businesses [21]. Engaging stakeholders throughout research and implementation processes ensures diverse perspectives are considered, enhancing the legitimacy and success of sustainability initiatives.

2. Decision-making tools in the context of circular economy

In the context of circular economy, decision-making tools are absolutely pivotal for effectively navigating the complex landscape of sustainable production and consumption. These tools, range from Assessment and Analytical Tools like Life Cycle Assessment (LCA), Material Flow Analysis (MFA), and Social Life Cycle Assessment (SLCA) to Strategic and Planning Tools such as the Circular Business Model Canvas and the Value Hill model, address the critical challenge of information asymmetry and foster a shared understanding across diverse organisational landscapes [17]. By quantifying circular practices' environmental, economic, and social impacts, these tools align strategic objectives and value propositions, simplifying the coordination and management of joint circular economy initiatives [2]. Furthermore, Operational and Process Optimization Tools, including Lean Management and Supply Chain Management techniques, streamline processes and enhance compatibility across organisational boundaries, addressing logistical complexities inherent in collaborative

projects [6]. Compliance and Reporting Tools, such as Environmental Management Systems (EMS) and the Global Reporting Initiative (GRI) framework, ensure that these efforts adhere to regulatory standards and sustainability reporting requirements, fostering transparency and building trust among stakeholders [22]. Innovation and Design Tools, including Cradle to Cradle design and Biomimicry, encourage collaborative innovation in product and process design, overcoming technological barriers and cultural differences to promote sustainable design principles [23]. At the same time, Stakeholder Engagement and Collaboration Tools are crucial for managing the diverse interests and expectations of various stakeholders, offering platforms for dialogue, decision-making, and knowledge sharing [7]. Lastly, Financial and Economic Analysis Tools like Cost-Benefit Analysis (CBA) and Return on Investment (ROI) analysis, address economic and financial misalignments by offering clear insights into the costs, savings, and potential returns of circular economy initiatives, supporting the development of shared investment models and risk-sharing mechanisms [24].

Despite the potential of decision-making tools to facilitate the circular economy, it's important to note that many of these tools were originally developed for linear economy models. These tools often rely on input from a single organisation, lacking comprehensive data across the product's extended lifecycle [1]. This limitation hampers their ability to address the circular economy's multi-stakeholder, systemic nature. Additionally, challenges such as data availability, competitive pressures, and the need for integration into existing systems further complicate their application [6, 8].

3. Requirements for Circular Economy

The workshop, arranged by the authors and attended by a global metal cutting tool manufacturer along with four SMEs customers that are users of, served as a platform to illuminate the primary operational challenges confronting the sector and to explore avenues for advancing towards a circular economy. During the workshop, the facilities ask questions about the challenges that SMEs face when it comes to refurbishing, repainting, extending the lifetime of the tool, and sorting the tools in recycling. First, facilitators developed a map on the whiteboard to visualise these challenges and their enablers. This was followed by an open discussion among the participants, which helped to clearly delineate the actors' roles and their contributions to overcoming these obstacles. (See [Table 1](#)). Insights into the roles of operators and management in promoting circular provided critical insights into the cutting tool industry's operational challenges, and circularity initiatives at manufacturing underscore the need for decision-making tools that facilitate communication and collaboration across organisational levels. Tools should support the integration of operational insights into strategic decision-making processes, ensuring that circularity efforts are aligned with organisational goals and effectively implemented on the ground. The framework should support network building and collaboration, providing platforms for sharing knowledge and resources to enhance industry circularity.

Table 1. Barriers and Opportunities for Collaborative Recycling Programs.

Workshop Finding	Implication for Decision-Making Tools	Enablers for Collaboration
Operational Practices and Recycling Efforts	Tools are needed to facilitate efficient material flow management and support recycling and scrap management initiatives.	Real-time tracking of material usage Scrap and recycling management modules
Benchmarking Across Companies	Incorporation of benchmarking functionalities to compare strategies and circular practices with industry standards and peers.	Benchmarking tools Industry standards database Peer comparison features
Roles of Operators and Management	Tools should enable seamless communication and collaboration across different organisational levels to integrate operational insights into strategic decision-making.	Role-based access and functionalities Communication platforms for sharing insights
Challenges and Opportunities in Circular Behaviors	Tools must offer comprehensive cost-benefit analysis features to assess the economic implications of circular practices.	Cost-benefit analysis modules Economic incentive identification tools
Fostering Collaboration for a Circular Economy	Facilitation of network building and collaboration among industry players to advance circular economy initiatives.	Collaboration platforms Resource and knowledge sharing features

4. Proposing a Requirement management model

The proposed requirements management model for decision-making in the circular economy delineates a structured approach towards fostering sustainable and effective practices within the circular economy landscape. This model comprises three interconnected stages, each characterised by detailed processes and information exchange to facilitate informed decision-making and drive the transition towards circularity.

In the initial pre-requirement collection stage, stakeholders engage in rigorous analysis of historical data, field best practices, and academic literature to generate preliminary assumptions about requirements and identify key actors involved in the circular economy transition. This stage involves the collection of diverse types of information, including historical trends in resource utilisation, case studies of successful circular economy initiatives, and academic research on sustainable practices. Stakeholders also identify key actors such as small businesses, technology providers, and policymakers, whose inputs are crucial for informing subsequent stages of the model. This stage involves the collection of primary data through surveys, interviews, and focus groups with stakeholders to gain insights into their unique perspectives and requirements. Additionally, stakeholders leverage secondary data sources such as academic literature, industry reports, and case studies to validate assumptions and benchmark their practices against industry standards. The outputs from this stage include comprehensive datasets containing stakeholder requirements, validated assumptions, and benchmarking insights, which serve as foundational elements for the subsequent modelling requirements stage. In the modelling requirements stage, stakeholders collaborate to model, balance, prioritise, and adjust requirements to derive optimal solutions, considering factors such as cost, feasibility, and stakeholder interests. This stage encompasses the synthesis and analysis of the information collected in previous stages to develop actionable insights and decision-making frameworks. Stakeholders utilise advanced modelling techniques such as scenario analysis, optimisation algorithms, and sensitivity analysis to evaluate

the potential impacts of different requirements and trade-offs. The outputs from this stage include decision support tools, scenario models, and optimisation frameworks that enable stakeholders to make informed decisions and prioritise actions towards achieving circularity in the economy. Throughout these stages, stakeholders play pivotal roles in stakeholder engagement, collaborative decision-making, and iterative refinement, thereby driving the transition towards circularity in the economy. By leveraging diverse sources of information and employing advanced analytical techniques, the proposed model empowers stakeholders to navigate the complexities of the circular economy landscape and implement sustainable practices that contribute to long-term environmental and economic sustainability.

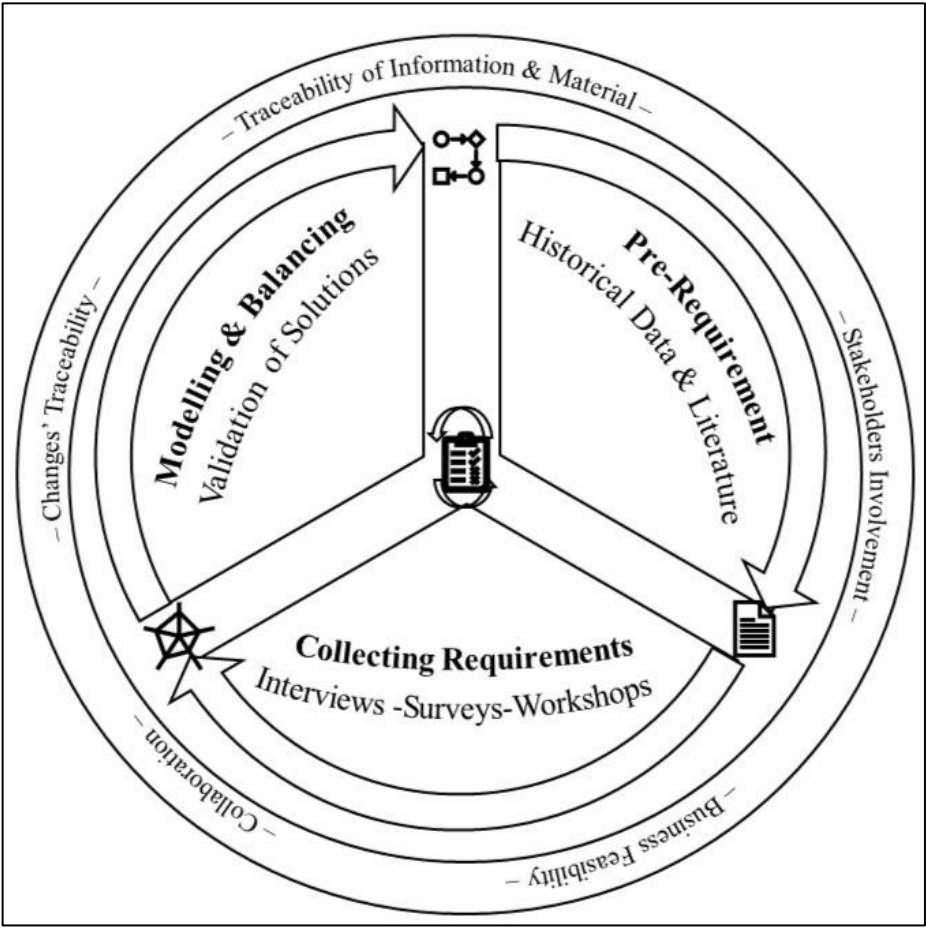


Figure 1. Framework for Developing decision tools to support collaboration for circular economy.

5. Discussion

5.1 Requirements for Circular Economy

Firstly, the emphasis on rigorous analysis of historical data, field best practices, and academic literature in the pre-requirement collection stage aligns with scholarly recommendations highlighting the importance of evidence-based decision-making and the incorporation of diverse perspectives in circular economy initiatives [17]. The model is based on empirical evidence and academic research, and tools are needed to facilitate efficient material flow management, recycling and scrap management initiatives. Enablers for collaboration, such as real-time tracking of material usage and dedicated modules for scrap and recycling management, address stakeholder requirements. The integration of stakeholder engagement frameworks and tools for environmental impact assessments in the pre-requirement collection stage aligns with inclusive decision-making and environmental considerations in circular economy strategies [2].

The model promotes transparency, accountability, and environmental stewardship by actively involving stakeholders in requirement elicitation and incorporating environmental considerations into decision-making frameworks. Furthermore, the model's focus on stakeholder collaboration, benchmarking functionalities, and cost-benefit analysis in the requirements management stage mirrors the literature's recommendations for promoting collaboration, sharing best practices, and assessing the economic viability of circular economy initiatives [17]. Through collaborative data collection and analysis, stakeholders can identify industry standards, assess the effectiveness of existing practices, and evaluate the financial implications of proposed initiatives, thereby fostering informed decision-making and resource allocation.

The model underscores the need for adaptability, scalability, and the integration of ethical and social considerations early in the development process, echoing literature on flexible decision-making frameworks and ethical dimensions in circular economy strategies [4, 11]. It acknowledges stakeholders' varied needs and values and aims for equitable, socially responsible circular economy outcomes. Additionally, by integrating cost-benefit analysis features and tools for identifying economic incentives, the model responds to challenges and opportunities in circular practices, enabling stakeholders to evaluate the economic impacts on decision-making and sustainable practice adoption. The model also emphasises collaboration within the circular economy, reflecting workshop insights on the importance of industry cooperation. Collaboration platforms, resource and knowledge-sharing features, and joint project management tools promote network building and collaborative efforts, advancing circular economy initiatives and encouraging a culture of cooperation and innovation in an interdisciplinary approach.

5.2 Interdisciplinary Collaboration in Circular Economy

The proposed model for creating decision-making tools in the circular economy aligns with insights from the literature review and responds to the challenges and opportunities presented by interdisciplinary collaboration for sustainable business practices. Interdisciplinary collaboration is recognised as crucial for addressing complex social-ecological system challenges and achieving sustainable outcomes [19]. By integrating diverse perspectives and adopting a systems-informed approach, the model ensures a comprehensive understanding of circular economy dynamics, thus facilitating more informed and effective decision-making processes. Furthermore, the challenges

associated with interdisciplinary research, such as coordinating across disparate disciplines and integrating multiple scales of analysis, present opportunities for creating innovative solutions [20]. The model prioritises the creation of collaborative platforms and tools that use common terminologies and frameworks to tackle challenges. By promoting effective communication and collaboration among stakeholders with different backgrounds and expertise, the model ensures that decision-making tools in the circular economy are technically strong, socially inclusive, and environmentally responsible.

6. Conclusion

To sum up, creating a circular economy requires a collaborative approach that goes beyond technical solutions to tackle important non-technical hurdles. This includes combining various scientific perspectives to create tools for decision-making that can adapt to different situations, consider ethical considerations, and support cooperation among stakeholders. The success of transitioning to a circular economy depends on addressing and overcoming challenges like regulations, aligning business models, and building trust among stakeholders.

References

- [1] M. Geissdoerfer, P. Savaget, N. M. P. Bocken, and E. J. Hultink, The Circular Economy – A new sustainability paradigm?, *Journal of Cleaner Production*, vol. 143, pp. 757-768, 2017/02/01/ 2017, doi: 10.1016/j.jclepro.2016.12.048.
- [2] M. Geissdoerfer, T. Santa-Maria, J. Kirchherr, and C. Pelzeter, Drivers and barriers for circular business model innovation, *Business Strategy and the Environment*, vol. 32, no. 6, pp. 3814-3832, 2023.
- [3] O. Khan, T. Daddi, H. Slabbinck, K. Kleinhans, D. Vazquez-Brust, and S. De Meester, Assessing the determinants of intentions and behaviors of organizations towards a circular economy for plastics, *Resour Conserv Recycl*, vol. 163, p. 105069, Dec 2020, doi: 10.1016/j.resconrec.2020.105069.
- [4] Y. Kalmykova, M. Sadagopan, and L. Rosado, Circular economy – From review of theories and practices to development of implementation tools, *Resources, Conservation and Recycling*, vol. 135, pp. 190-201, 2018/08/01/ 2018, doi: 10.1016/j.resconrec.2017.10.034.
- [5] J. S. Mboli, D. Thakker, and J. L. Mishra, An Internet of Things-enabled decision support system for circular economy business model, (in English), *Software: Practice and Experience*, Article vol. 52, no. 3, pp. 772-787, Mar 2020, doi: 10.1002/spe.2825.
- [6] S. Seuring and M. Müller, From a literature review to a conceptual framework for sustainable supply chain management, *Journal of Cleaner Production*, 2008, vol. 16, no. 15, pp. 1699-1710, 2008/10/01/ doi: 10.1016/j.jclepro.2008.04.020.
- [7] M. S. Reed *et al.*, Who's in and why? A typology of stakeholder analysis methods for natural resource management, *J Environ Manage*, vol. 90, no. 5, pp. 1933-49, Apr 2009, doi: 10.1016/j.jenvman.2009.01.001.
- [8] M. Antikainen, T. Uusitalo, and P. Kivikytö-Reponen, Digitalisation as an enabler of circular economy, *Procedia CIRP*, 2018, vol. 73, pp. 45-49.
- [9] C. Ansell and A. Gash, Collaborative Governance in Theory and Practice, *Journal of Public Administration Research and Theory*, 2008, vol. 18, no. 4, pp. 543-571, doi: 10.1093/jopart/mum032.
- [10] P. Zaraté, *Tools for Collaborative Decision-Making*, Wiley, Hoboken, 2013.
- [11] Y. Cai, F. Jin, J. Liu, L. Zhou, and Z. Tao, A survey of collaborative decision-making: Bibliometrics, preliminaries, methodologies, applications and future directions, *Engineering Applications of Artificial Intelligence*, 2023, vol. 122, p. 106064.

- [12] S. Peck, Group Model Building: Facilitating Team Learning Using System Dynamics, *Journal of the Operational Research Society*, vol. 49, no. 7, pp. 766-767, 1998/07/01 2017, doi: 10.1057/palgrave.jors.2600567.
- [13] M. Marttunen, J. Lienert, and V. Belton, Structuring problems for Multi-Criteria Decision Analysis in practice: A literature review of method combinations, *European journal of operational research*, 2017, vol. 263, no. 1, pp. 1-17.
- [14] J. McAvoy, D. Sammon, and I. Owens, A simple tool to assist in agile methodology adoption decisions, *Journal of decision systems*, 2007, vol. 16, no. 4, pp. 451-468.
- [15] K. E. Maani and V. Maharaj, Links between systems thinking and complex decision making, *System Dynamics Review: The Journal of the System Dynamics Society*, 2004, vol. 20, no. 1, pp. 21-48.
- [16] S. Diamond, S. Szigeti, and A. Jofre, Building tools for creative data exploration: a comparative overview of data-driven design and user-centered design, in *Distributed, Ambient and Pervasive Interactions: 5th International Conference, DAPI 2017, Held as Part of HCI International 2017, Vancouver, BC, Canada, July 9–14, 2017, Proceedings 5*, 2017: Springer, pp. 514-527.
- [17] N. M. P. Bocken, I. de Pauw, C. Bakker, and B. van der Grinten, Product design and business model strategies for a circular economy, *Journal of Industrial and Production Engineering*, vol. 33, no. 5, pp. 308-320, 2016/07/03 2016, doi: 10.1080/21681015.2016.1172124.
- [18] W. Akkalatham and A. Taghipour, Pro-environmental behavior model creating circular economy in steel recycling market, empirical study in Thailand, *Environmental Challenges*, vol. 4, 2021/8// 2021, doi: 10.1016/j.envc.2021.100112.
- [19] N. G. Pricope *et al.*, Addressing integration challenges of interdisciplinary research in social-ecological systems, *Society & Natural Resources*, 2020, vol. 33, no. 3, pp. 418-431.
- [20] K. L. Daniel, M. McConnell, A. Schuchardt, and M. E. Pfeffer, Challenges facing interdisciplinary researchers: Findings from a professional development workshop, *PLOS ONE*, 2022, vol. 17, no. 4, p. e0267234, doi: 10.1371/journal.pone.0267234.
- [21] R. Freeth and G. Caniglia, Learning to collaborate while collaborating: advancing interdisciplinary sustainability research, *Sustainability science*, 2020, vol. 15, no. 1, pp. 247-261.
- [22] S. A. Melnyk, R. P. Sroufe, and R. Calantone, Assessing the impact of environmental management systems on corporate and environmental performance, *Journal of Operations Management*, vol. 21, no. 3, pp. 329-351, 2003/05/01/ 2002, doi: 10.1016/s0272-6963(02)00109-2.
- [23] W. McDonough, M. Braungart, and T. H. Kerry, *The Hannover principles: Design for sustainability*. W. McDonough Architects, 2003.
- [24] P. Ghisellini, C. Cialani, and S. Ulgiati, A review on circular economy: the expected transition to a balanced interplay of environmental and economic systems, *Journal of Cleaner Production*, vol. 114, pp. 11-32, Feb 15 2016, doi: 10.1016/j.jclepro.2015.09.007.